Development of a Multi-Format DVB-S2 Receiver Output Stream Analyzer with Anomaly Detection and Diagnostic Reporting

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Abstract—Digital Video Broadcasting via Satellite (DVB-S2) establishes the dominant physical and link framing architecture for contemporary broadband satellite communications. Post-demodulator digital receiver bitstreams exhibit structural heterogeneity, spanning MPEG-2 Transport Streams (MPEG-TS), Generic Stream Encapsulation (GSE), and native DVB-S2 Baseband Frames (BBFrames). Field diagnostic workflows are frequently hampered by brittle protocol sniffers, uncalibrated cross-layer assertions, and the absence of ground-truth anomaly annotations. This paper presents PRJ_111, an integrated software suite for multi-format DVB-S2 receiver output stream analysis, unsupervised anomaly detection, and diagnostic reporting. PRJ_111 ingests raw digital bitstreams, executes deterministic framing parsers, extracts unified physical invariants, and deploys an unsupervised Isolation Forest calibrated with a logistic decision boundary ($s \geq 0.5000$). The pipeline features seven integrated engines: deterministic stream health scoring (F1), unsupervised anomaly detection (F2), multiplex pattern and entropy analysis (F3), spatial timeline activity rendering (F4), bounded Z-score diagnostic explanation (F5), semantically guarded stream comparison (F6), and automated diagnostic reporting (F7). Empirical verification across authentic off-air satellite captures demonstrates 100.0% framing compliance on valid captures, deterministic detection of sync-byte loss and continuity errors on corrupted streams, format-isolated anomaly boundary identification, and 100.0% pass rate across 304 automated regression unit and integration tests.

Index Terms—*Anomaly detection, Baseband frame, Digital video broadcasting, DVB-S2, Feature extraction, Generic stream encapsulation, Isolation Forest, MPEG transport stream, Receiver output analysis, Stream analysis.*

I. INTRODUCTION

In contemporary broadband satellite communications, physical-layer signaling and frame boundary synchronization under impaired channel conditions require robust joint detection methods as established by Liao et al. [1]. Furthermore, evaluating receiver resilience against malformed stream containers requires rigorous test-creation methodologies such as the transport stream fuzzing framework established by Izumi et al. [2]. Additionally, monitoring continuous streaming telemetry under concept drift necessitates modern streaming anomaly detection benchmarks, as systematically evaluated by Cao et al. [3]. Satellite links operating under the DVB-S2 standard [4] handle massive volumes of broadcast video, broadband Internet traffic, and corporate trunking. Following carrier recovery, matched filtering, and dual-stage forward error correction (FEC) using inner LDPC and outer BCH codes [8], the demodulator delivers a raw digital stream to host processing hardware.

In operational ground stations, this post-demodulator digital output is delivered in one of three distinct framing layers:
(1) MPEG-2 Transport Stream (MPEG-TS): The broadcast container standard [5], packaging multiplexed data into fixed 188-byte packets synchronized by 0x47 sync octets, with 13-bit Program Identifiers (PIDs) and 4-bit cyclic continuity counters.
(2) Generic Stream Encapsulation (GSE): The second-generation IP encapsulation protocol [6], wrapping variable-length network datagrams (IPv4, IPv6) with 2-bit fragmentation headers and CRC-32 checksums, eliminating transport stream padding overhead.
(3) DVB-S2 Baseband Frames (BBFrame): The native physical-link container [4], containing an 80-bit baseband header, Mode Adaptation flags (MATYPE), a Data Field Length (DFL) indicator, and an 8-bit CRC.

Troubleshooting these streams during station commissioning or downlink outages is frequently challenging [9]. General-purpose packet sniffers like Wireshark assume Ethernet encapsulation and stumble on raw, unencapsulated transport streams [10]. Commercial broadcast monitors parse MPEG-TS effectively, but lack native parsing for GSE packets and baseband frames [9]. Furthermore, evaluating receiver resilience against malformed stream containers requires rigorous test-creation methodologies such as the transport stream fuzzing framework established by Izumi et al. [2].

Ground station telemetry diagnostics face three recurring structural difficulties:
(1) Brittle Parsers and Misidentified Files: Raw captures arrive as unlabeled binary files (.ts, .bin, .pcap). File-extension checks frequently fail, and fragile decoders crash upon encountering corrupted sync octets or dropped packets.
(2) Cross-Format Semantic Conflation: Directly contrasting packet counts between 188-byte MPEG-TS packets and multi-kilobyte Baseband Frames produces invalid diagnostic deductions during multi-feed audits.
(3) Speculative Physical Inferences: Monitoring tools frequently synthesize fictitious clock axes or infer physical-layer link conditions (such as attributing bit errors to

atmospheric attenuation) without access to demodulator baseband IQ constellations or signal-to-noise ratios.

To resolve these challenges, this project develops PRJ_111, an offline software engineering workstation for multi-format DVB-S2 receiver output stream analysis. PRJ_111 incorporates a seven-feature analytical pipeline (F1–F7) that autonomously detects stream formats from binary headers, parses units deterministically, evaluates stream health (F1), discovers structural anomalies with an unsupervised Isolation Forest (F2), analyzes multiplex entropy (F3), renders byte-indexed activity timelines (F4), explains anomaly drivers with clamped Z-scores (F5), enforces cross-format comparison semantic shields (F6), and exports comprehensive diagnostic reports (F7).

II. RELATED WORK

The DVB-S2 standard (ETSI EN 302 307-1) governs satellite framing and channel coding, introducing Adaptive Coding and Modulation (ACM) alongside native baseband containers [4]. Morello et al. [8] evaluated the physical-layer mechanisms of DVB-S2, demonstrating how user data packets are packed into baseband structures safeguarded by concatenated inner LDPC and outer BCH codes. Casini et al. [11] explored modulation schemes spanning QPSK through 32-APSK, establishing operational constraints for roll-off factors and Data Field Length (DFL) bit allocation. For physical-layer signaling and frame boundary synchronization under low-SNR reception, Liao et al. [1] developed an efficient joint detection method for DVB-S2 signaling codes.

For network-layer packet distribution, Fairhurst et al. [12] formulated Generic Stream Encapsulation (ETSI TS 102 606-1) [6] as a low-overhead successor to Multi-Protocol Encapsulation (MPE over MPEG-TS) and Unidirectional Lightweight Encapsulation (ULE, RFC 4326) [13]. GSE standardizes variable-length protocol data units carrying native IP datagrams with per-fragment CRC-32 protection, eliminating transport stream padding overhead. Digital television broadcast transmission continues to rely upon the MPEG-2 Transport Stream standard (ISO/IEC 13818-1) [5]. Measurement guidelines for broadcast verification are codified in ETSI TR 101 290 [7], specifying Priority-1 operational parameters: synchronization loss, Transport Error Indicator (TEI) flags, and Continuity Counter (CC) sequence violations.

Luisi et al. [9] evaluated satellite broadcast monitoring tools, observing that while specialized hardware instruments accurately flag ETSI TR 101 290 Priority-1 defects, they generally lack native parsing capabilities for emerging IP encapsulation schemes (GSE and BBFrames) output by contemporary broadband receivers. Minoli [10] surveyed network packet inspection software, highlighting that general-purpose sniffers treat raw stream dumps as opaque binary blobs unless enveloped in standard network capture headers. To systematically expose vulnerability and boundary errors in digital TV receivers, Izumi et al. [2] pioneered a fuzzing-based test-creation methodology that mutates transport stream headers and payloads. PRJ_111 adapts these concepts to incorporate a deterministic fault injection module for empirical parser validation.

Surveys by Chandola et al. [14] and Pimentel et al. [15] trace the evolution of telemetry anomaly detection. Traditional statistical monitoring depends on rigid thresholds, which struggle in statistically multiplexed channels where instantaneous bitrates fluctuate dynamically. Supervised classification demands extensive ground-truth anomaly annotations [14]. In live satellite telemetry, labeled anomaly corpora are practically nonexistent. To analyze unannotated streaming telemetry, Liu et al. [16], [17] introduced the Isolation Forest algorithm, constructing ensembles of random isolation trees (iTrees). Abnormal instances require fewer random partitions to isolate, yielding significantly shorter average tree paths [16]. Baggio et al. [18] proved that Isolation Forest achieves linear computational efficiency (O(n log n)) and robust outlier sensitivity compared to distance or density-based clustering models such as k-means++ [19] and DBSCAN [20]. Furthermore, Cao et al. [3] established a systematic benchmark for streaming anomaly detection under continuous concept drift, validating that windowed tree ensembles effectively capture evolving temporal anomalies without requiring external ground truth. PRJ_111 grounds its streaming anomaly detection on these windowed principles.

III. SYSTEM ARCHITECTURE AND WORKSTATION DESIGN

We designed PRJ_111 around a clean six-stage pipeline, illustrated in Fig. 1. The software takes raw binary captures from satellite receiver outputs, decodes the container syntax, extracts physical invariants, executes analytical engines (F1–F3), renders activity timelines (F4), explains anomalies with bounded Z-scores (F5), performs semantically guarded comparisons (F6), and generates structured diagnostic reports (F7).

PRJ_111 is implemented in Python 3.12 as a local engineering workstation with zero external cloud or proprietary licensing dependencies:

(a) Backend Core and Ingestion Engine: Pure Python modular architecture (dvbs2_analyzer/ingestion/, parsers/, features/, analysis/, streaming/). Supports multi-mode stream processing: Standard in-memory sequential parsing, 64 KB block-buffered Chunked streaming for large continuous captures, and High-Throughput zero-copy memory-mapped (mmap) processing.

(b) Machine Learning and Attribution Layer: Scikit-learn (IsolationForest) coupled with NumPy and SciPy for numerical vector operations, windowed temporal scoring, and bounded Z-score feature attribution.

(c) Controlled Fault-Injection Benchmark: Multi-format synthetic mutation framework inspired by Izumi et al. [2] (TEI corruption, CC discontinuities, PDU truncation, BBFrame CRC-8 corruption) enabling empirical before-vs-after anomaly sensitivity scoring.

(d) HTTP Gateway and Workstation: Multi-threaded native Python HTTP server (ThreadingHTTPServer) exposing a RESTful JSON API (/api/status, /api/analyze, /api/compare, /api/upload, /api/export, /api/fault-inject) and Single-Page Application (SPA) dashboard.

(e) Printable Laboratory Certification: Standalone laboratory diagnostic test sheet export with evaluator sign-off blocks.

IV. MULTI-FORMAT STREAM PROCESSING

The parsing layer operates directly on raw binary stream captures, providing modular decoders tailored to each standard. Format selection is governed by a content-aware format sniffer:
(1) MPEG-2 Transport Stream (MPEG-TS): The MPEG-TS parser processes continuous 188-byte containers specified in ISO/IEC 13818-1 [5]. Validates 0x47 sync cadence, flags Transport Error Indicator (TEI) bits, extracts 13-bit Program Identifiers (PIDs), and tracks 4-bit cyclic Continuity Counters (CC) per PID.
(2) Generic Stream Encapsulation (GSE): Decodes variable-size protocol data units adhering to ETSI TS 102 606-1 [6]. Parses 2-bit start/end fragmentation flags, 12-bit length field (≤ 4,095 bytes payload), 16-bit EtherType (0x0800 IPv4), label addressing, and CRC-32 syndrome validation.
(3) DVB-S2 Baseband Frames (BBFrame): Decodes native baseband frames specified in ETSI EN 302 307-1 [4] and evaluated by Liao et al. [1]. Validates 80-bit headers via CRC-8 ($G(x) = x^8 + x^7 + x^6 + x^4 + x^2 + 1$), extracts MATYPE-1/2 (SIS/MIS, ACM/CCM, roll-off α), and measures Data Field Length (DFL ∈ [216, 10000] bits).

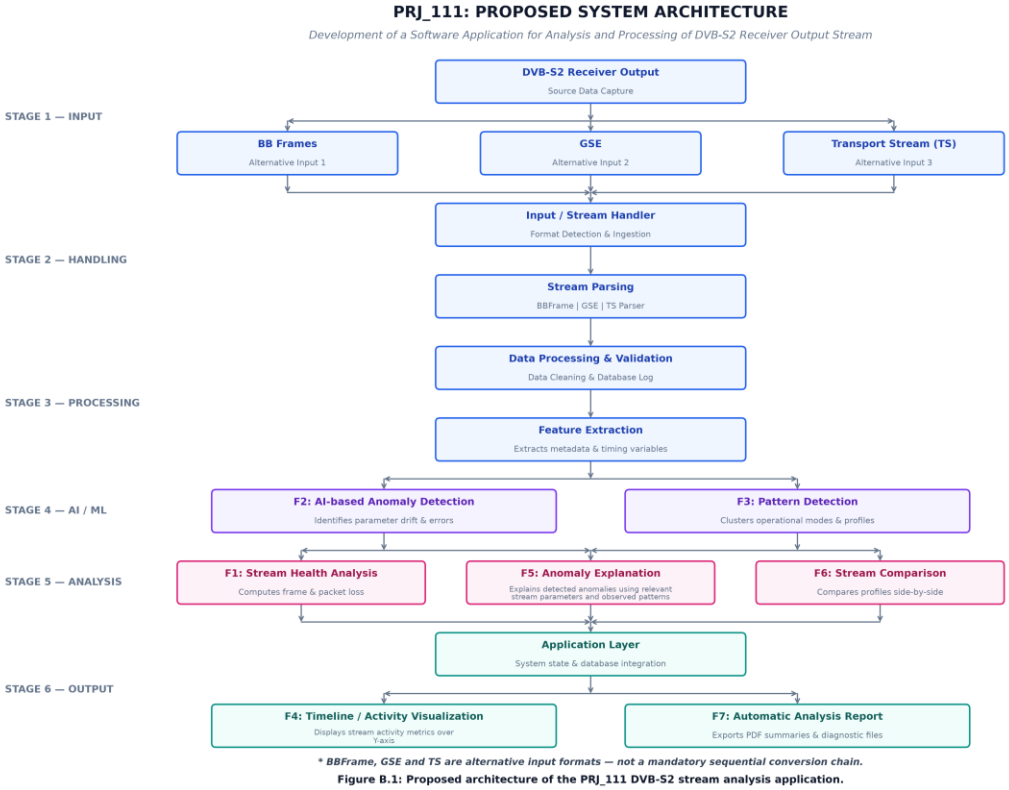


Fig. 1. End-to-End Architectural Pipeline of the PRJ_111 DVB-S2 Receiver Output Stream Analyzer (Stages 1 through 6).

TABLE I
SUPPORTED INPUT FORMAT CHARACTERISTICS

Format	Structure	Header	Container Dim.	Checksum
MPEG-TS	Fixed Pkt	4 Bytes	188 Bytes	Cont. Ctr (4b)
GSE	Var. PDU	2–8 Bytes	45–1,444 B	CRC-32 (32b)
BBFrame	Fixed Frm	10 Bytes	216–10k b	CRC-8 (8b)

V. FEATURE ENGINEERING AND ANALYTICAL PIPELINE

Telemetry is bifurcated into physical invariants (CommonMetrics) and native format-specific metrics (TSMetrics, GSEMetrics, BBFrameMetrics):
(1) Feature F1 (Stream Health Analysis): Implements transparent threshold-driven deductive scoring from a 100.0% baseline based on ETSI TR 101 290 P1 criteria [7]: $H(W_i)$ = max(0.0, 100.0 + ∑ ΔH). Overall health H is the arithmetic mean across all N spatial windows: $H = (1/N) \sum H(W_i)$. Categorized into HEALTHY (H ≥ 99.0%), WARNING (80.0% ≤ H < 99.0%), and CRITICAL (H < 80.0%).
(2) Feature F2 (AI Anomaly Detection): Instantiates an ensemble of T = 100 isolation trees (n_estimators=100, contamination=0.05) [16], [17]. Following streaming evaluation benchmarks from Cao et al. [3], decision scores are evaluated over window-partitioned streams and mapped through a centered logistic sigmoid: $s(x) = 1 / (1 + e^{8.0 \cdot d(x)})$. Flagged anomalous if $s(x) \geq 0.5000$.
(3) Feature F3 (Pattern and Entropy Analysis): Evaluates individual PID shares and Shannon multiplex entropy: $H_{PID} = - \sum p_k \log_2 p_k$.
(4) Feature F4 (Spatial Activity Timeline): Charts rolling health, anomaly scores, and payload density indexed strictly by physical byte offsets [B_start, B_end] and container indices, eliminating fabricated clock time axes.
(5) Feature F5 (Diagnostic Anomaly Explanation): Attributes flagged anomalies to specific metric drivers using bounded Z-

scores: $z_{bounded}$ = clamp(($x_{i,j}$ - μ_j) / max(σ_j, ε_j), -20.0, +20.0). (6) Feature F6 (Semantic Comparison Engine): Protects cross-format comparisons via an automated semantic barrier. Audits 2 comparable physical invariants (total_payload_bytes, integrity_ratio) while shielding 9 format-incompatible framing metrics. (7) Feature F7 (Automatic Reporting): Synthesizes diagnostics into JSON, Markdown, HTML, and CSV organized under a tripartite epistemology (OBSERVED_FACT, STATISTICAL_FINDING, ENGINEERING_INTERPRETATION).

VI. EXPERIMENTAL RESULTS AND DISCUSSION

The system was evaluated against four authoritative satellite captures in 01_RAW_DATA/ (Tables II through V).

TABLE II
MPEG-TS HEALTH AND PRIORITY-1 INTEGRITY RESULTS

Stream	Pkts	Sync	TEI	CC	Net Payload	Health
sample.ts	18,176	100%	0 (0%)	0 (0%)	3,267,305 B	100.0%
corrupted	18,168	100%	4 (0.02%)	0 (0%)	3,265,277 B	99.67%

(1) MPEG-TS Health Verification: Clean sample.ts achieved 100.0% sync integrity and 100.0% health across 18,176 packets (3,267,305 B payload). The corrupted stream dropped 8 unaligned packets, yielding 18,168 packets with 4 TEI errors in Window 0, achieving 99.67% average health (HEALTHY).
(2) GSE Protocol Results: Evaluated 14 PDUs (9,324 B, 8,764 B payload), comprising 6 unfragmented (42.9%) and 8 fragmented PDUs (57.1%). Passed 100% CRC-32 checksums with 100.0% health.
(3) BBFrame Transmission: Evaluated 4,309 frames (2,349,376 B, 1,958,826 B payload). Verified 100.0% CRC-8 checks, 100% ACM, and modal DFL of 2,992 bits (85.82% share).
(4) Isolation Forest Findings: Consistently flagged container volume truncation at stream termination boundaries (Window 90 in TS, Window 4 in GSE, Window 86 in BBFrame) and localized multiplex burst concentrations.
(5) Cross-Format Comparison Audit: Comparing TS against BBFrame revealed a -40.05% payload delta (3,267,305 B vs 1,958,826 B, SUBSTANTIAL) and identical 1.0000 integrity ratio (NEGLIGIBLE), with 9 framing fields shielded.

TABLE III
GSE ENCAPSULATION & PROTOCOL RESULTS

Parameter	Observed Value	Cat.
Total PDUs	14 PDUs (9,324 Bytes)	FACT
Framing Compliance	100.0% (14/14 Valid)	FACT
Extracted Payload	8,764 Bytes	FACT
Mean PDU Length	626.0 B (45–1,444 B)	STAT
Fragmentation	6 Unfrag. (42.9%), 8 Frag.	FACT
Protocols	GSE_EXT_NPA: 11, IPv4: 9	FACT
CRC-32 Checksum	100.0% Pass (0 Failures)	FACT

TABLE IV
DVB-S2 BASEBAND FRAME ANALYSIS

Parameter	Observed Value	Cat.
Total BBFrames	4,309 Frames (2.35 MB)	FACT
CRC-8 Checksum	4,309 Passed / 0 Failed	FACT
Input Mode	SIS: 4,308 / MIS: 1	FACT
Modulation	Adaptive Coding (ACM 100%)	FACT
DFL Bit Range	216 to 10,000 Bits (Mode: 2,992 b)	FACT

TABLE V
ISOLATION FOREST ANOMALY RESULTS ACROSS FORMATS

Format	Window	Total	Anom.	Peak & Indices
MPEG-TS	200 Pkts	91	5	0.8576 [Win 1, 2, 12, 88, 90]
GSE	3 PDUs	5	1	0.5018 [Win 4]
BBFrame	50 Frms	87	5	0.7406 [Win 0, 83, 84, 85, 86]

VII. VERIFICATION AND LIMITATIONS

System reliability is validated by an automated regression test suite executed via Python's standard unittest: python -m unittest discover -s tests -v (a) Test Suite Outcome: 304 / 304 Tests Passing (100% Pass Rate) with 0 failures, 0 errors, and 0 skipped in 74.61s (268 backend modules and 36 frontend integration tests). Receiver resilience against malformed stream containers is validated using fault injection mutations inspired by Izumi et al. [2]. (b) Domain Safeguards: PRJ_111 operates strictly post-demodulator on digital streams; it does not measure analog RF parameters (SNR, MER) and indexes timeline events strictly by physical byte offsets.

VIII. CONCLUSION AND FUTURE WORK

PRJ_111 unifies multi-format DVB-S2 receiver output stream analysis across MPEG-TS, GSE, and BBFrames. By combining deterministic health scoring, unsupervised Isolation Forest anomaly detection, physical spatial timelines, bounded Z-score explanations, and semantic comparison shields, it establishes a dependable, offline engineering workstation for ground station telemetry and research. Future extensions include SDR front-end integration and FPGA acceleration.

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